DECK 05 · WEEK 3 · MORNING 1

Macro Policy, Geopolitics, Trade & FX

Six CFA modules on the macro forces that drive every cross-border price

CFA Level I · Vol 2 Learning Modules 3–8

STUDIO TONTON GUG · Finance Fundamentals

Roadmap — Macro Policy, Geopolitics, Trade & FX

6 sections · ~2 hours · Built bottom-up from CFA readings

01 **Fiscal policy toolkit + multipliers**
Roles · automatic stabilizers · multiplier $1/[1-c(1-t)]$ · five fiscal tools · debt dynamics · three lags

02 **Monetary policy transmission**
OMO · policy rate · reserve requirements · 4 channels · inflation targeting · the neutral rate

03 **Geopolitics for investors**
State vs non-state actors · autarky / hegemony / multilateralism / bilateralism · event, exogenous, thematic risk

04 **Trade theory + the trade balance**
Comparative advantage · tariff welfare and deadweight loss · five bloc types · $X-M = (S-I)+(T-G)$

05 **FX market structure & quoting**
Turnover and instrument mix · price/base convention · bid-offer and inversion · the IMF eight regimes

06 **FX arithmetic: cross-rates, IRP, PPP**
Real FX · cross-rate chain · triangular arb · covered IRP and forward points · PPP as a long-run anchor

Learning objectives

By the end of this deck you should be able to:

1. Compute the fiscal multiplier $1/[1 - c(1 - t)]$ for a given MPC and tax rate, explain the tax rate as the leakage, and distinguish it from the money multiplier.

2. Describe the three monetary-policy tools and the four transmission channels, and build the neutral rate from trend growth plus the inflation target.

3. Place a country on the four-archetype framework (autarky, hegemony, multilateralism, bilateralism) and classify a risk as event, exogenous, or thematic.

4. Apply comparative advantage from opportunity costs, and decompose a tariff into consumer surplus, producer surplus, revenue, and deadweight loss.

5. Read an FX quote in price/base terms, invert a two-sided quote correctly, and derive a cross-rate through the USD — inverting a leg where required.

6. Compute a forward rate and forward points from spot and two interest rates, and identify the arbitrage direction when a dealer misquotes the forward.

SECTION 01

Fiscal policy toolkit

Fiscal policy uses government spending and taxation to steer aggregate demand, redistribute income, and allocate resources across sectors

The curriculum states one shared goal for both fiscal and monetary policy: stable positive growth with stable low inflation. Everything below is a means to that end.

ROLE 1 · DEMAND MANAGEMENT

Raise G or cut T when output is below potential; do the reverse to cool an overheating economy. The deficit is the instrument, not the objective.

ROLE 2 · REDISTRIBUTION

Progressive income taxes and targeted transfers compress the income distribution; indirect taxes on spending tend to widen it. A political choice, not a macro one.

ROLE 3 · ALLOCATION

Subsidies, excise duties and infrastructure spending steer resources toward chosen sectors — externality correction, or industrial policy under another name.

AUTOMATIC STABILIZERS · AND THE ARGUMENT THAT NEVER ENDS

Progressive taxes and unemployment benefits cushion a shock with no policy decision at all — the deficit widens as GDP falls and narrows as it recovers. The curriculum is explicit that their great advantage is exactly this: they are automatic, so they escape every implementation lag. Discretionary policy does not. Keynesians argue stimulus is powerful when there is slack; the opposing view is that it is temporary and monetary policy is the better stabilization tool. Because automatic stabilizers move the deficit on their own, economists strip them out and look at the STRUCTURAL budget deficit to read a government's true fiscal stance.

Source: CFA Institute, Curriculum Vol 2, LM 3 §1 "Introduction to Monetary and Fiscal Policy" and "Roles and Objectives of Fiscal Policy", pp. 80–88.

The fiscal multiplier turns one unit of government spending into a larger change in output — and the tax rate is the leakage that shrinks it

The formula assumes idle capacity. At full employment extra G bids up prices and crowds out private investment rather than adding real output.

$$\text{Multiplier} = 1 / [\, 1 - c \times (1 - t) \,]$$

where: c = marginal propensity to consume out of DISPOSABLE income; t = marginal tax rate. With no tax the expression collapses to the familiar 1/(1 − c).

FISCAL STIMULUS — WHAT THE TAX RATE COSTS YOU

A GBP 10 bn infrastructure package. MPC out of disposable income c = 0.9, marginal tax rate t = 0.2. With no tax at all the multiplier would be 1/(1 − 0.9) = 10. What does tax do to that?

1. Tax is the leakage: each extra GBP 1 of income leaves only (1 − t) = GBP 0.80 as disposable income.
2. Effective propensity to consume out of pre-tax income = c(1 − t) = 0.9 × 0.8 = 0.72.
3. Multiplier = 1 / (1 − 0.72) = 1 / 0.28 = 3.57 — a 20% tax rate alone cuts 10 down to 3.57.
4. ΔY = 3.57 × GBP 10 bn = GBP 35.7 bn of additional aggregate output.

Multiplier = 3.57 → ΔY ≈ GBP 35.7 bn, provided there is slack in the economy

[1] Balanced-budget multiplier: raise G and taxes by the same amount and output still rises, because consumption falls only by c × ΔT. Its value is always exactly one.

Source: CFA Institute, Curriculum Vol 2, LM 3 §2 "Fiscal Policy Tools" — the fiscal multiplier and the balanced budget multiplier, pp. 96–97.

The curriculum lists five fiscal tools, and they differ far more in how fast you can deploy them than in how hard they hit

Indirect taxes can be changed almost the moment they are announced; direct taxes usually need a year of notice; capital projects need a parliament and a planning system.

Fiscal tool	Route to demand	Speed	Main limitation
Current government spending	Direct, full value	Months	Hard to reverse once legislated
Capital expenditure	Direct + supply side	1–3 years	Planning and procurement lags
Transfer payments	Indirect — via MPC	Weeks	Leaks into saving; hard to undo
Direct taxes (income)	Indirect — disposable Y	Up to a year	Weak if the cut is saved
Indirect taxes (VAT, excise)	Indirect — via prices	Almost immediate	Regressive; distorts prices

THREE IMPLEMENTATION LAGS · RECOGNITION → ACTION → IMPACT

RECOGNITION lag — the data arrive late and get revised, so it takes time to even see the shock. ACTION lag — legislatures must debate and pass the measure. IMPACT lag — households and firms then take time to respond. Stack the three and discretionary stimulus routinely lands after the shock it was written for has passed, which is the strongest single argument for leaning on automatic stabilizers instead.

[1] Capital expenditure is the odd one out: unlike a change in personal or indirect taxes, it can raise the productive potential of the economy, not just current demand.

Source: CFA Institute, Curriculum Vol 2, LM 3 §2 "Fiscal Policy Tools", pp. 92–96; §3 on implementation lags, p. 100.

Debt is sustainable when real growth outruns the real interest rate on the debt — which is why the interest bill tells you more than the debt stock

Read the last two columns together. Japan carries 237% of GDP in debt and pays 0.4% of GDP to service it; the US carries half the debt and pays seven times the interest. The rate is the story.

$$\Delta\,(\text{Debt} / \text{GDP}) \approx (r - g) \times (\text{Debt} / \text{GDP}) - \text{Primary surplus} / \text{GDP}$$

where: r = real interest rate on the debt; g = real GDP growth. The curriculum states this verbally: if real growth is below the real rate on the debt, the ratio worsens even while the economy grows.

General government	Debt/GDP 1995	Debt/GDP 2015	Δ (p.p.)	Net interest, % GDP 2015
Japan	94.7%	237.4%	+142.7	0.4%
United States	83.2%	125.3%	+42.1	2.8%
United Kingdom	51.4%	111.7%	+60.3	2.0%
Germany	54.1%	78.9%	+24.8	0.9%
Australia	57.3%	64.1%	+6.8	0.3%
OECD average	65.8%	85.3%	+19.5	1.9%

[1] Ricardian equivalence: if households anticipate the taxes needed to repay debt, they save the cut and the multiplier goes to zero.

Source: CFA Institute, Curriculum Vol 2, LM 3 §1 "Deficits and the National Debt", pp. 88–90, Exhibits 7 and 8 (OECD general government data).

SECTION 02

Monetary policy transmission

Central banks influence money and credit with three tools — open market operations, the official policy rate, and reserve requirements

Open market operations are the day-to-day workhorse. Reserve requirements are barely touched in the US or the euro area — the Bank of England no longer even sets a minimum — but remain an active lever in China.

Tool	Mechanism	In practice
Open market operations	Buy/sell govt securities to inject or drain reserves; broad money follows via the multiplier	Fed Open Market Desk
Policy / refinancing rate	Rate at which the central bank lends short-term to banks — anchors the front of the curve	BoE Bank Rate, ECB MRO
Reserve requirements	Minimum share of deposits banks must hold — a higher ratio shrinks lending capacity	PBoC RRR

BEYOND THE THREE · QE

Large-scale asset purchases push down long rates once the policy rate is at its floor. Balance sheets at the Fed, ECB and BoJ expanded several-fold after 2008.

BEYOND THE THREE · NEGATIVE RATES

The ECB (2014–22) and BoJ (2016–24) charged banks on excess reserves. Effective at the margin, corrosive for bank margins if held too long.

BEYOND THE THREE · GUIDANCE

“Rates stay here until inflation is durably at target.” Moves the expected path without moving today’s rate — it works only if the bank is believed.

[1] MONEY multiplier: how far base money expands into broad money, governed by the reserve requirement. A monetary quantity.

[2] FISCAL multiplier: $1/[1 - c(1 - t)]$, the output response to a spending change. Same word, unrelated mechanism — do not conflate them.

Source: CFA Institute, Curriculum Vol 2, LM 4 §1 "Monetary Policy Tools and the Monetary Transmission Mechanism", pp. 110–113.

The policy rate reaches the real economy through four interconnected channels — short rates, asset prices, the exchange rate, and expectations

The curriculum stresses these are interconnected, not parallel — and that full transmission takes many months. That lag is precisely why central banks must act on forecast inflation, never on realized inflation.

1

Short-term interest rates

Policy rate → interbank and bank base rates → household and corporate borrowing costs → consumption and investment. The most direct leg, and the fastest.

2

Asset prices

A higher discount rate lowers bond, equity and real-estate prices and the value of capital projects. Falling wealth then feeds back into consumption growth.

3

Exchange rate

A higher domestic rate tends to appreciate the currency: exports get dearer abroad, imports cheaper at home — so import prices pull measured inflation down.

4

Expectations / confidence

Firms and households extrapolate. If the market reads one hike as the start of a series, borrowing and asset prices adjust before any further move is made.

Source: CFA Institute, Curriculum Vol 2, LM 4 §1, Exhibit 3 "A Stylized Representation of the Monetary Transmission Mechanism" (after the Bank of England), pp. 113–114.

Inflation targeting anchors expectations by committing the central bank to a public numerical goal — and the curriculum names only two major holdouts

New Zealand pioneered the framework: announced in 1988, given legal force by the Reserve Bank of New Zealand Act 1989, with an original range of 0–2%. Chile, Canada, Israel and the UK had copied it within three years.

WHY 2% · NOT 0%

Aiming at zero risks overshooting into deflation, which monetary policy handles badly. A small positive target buys a buffer, lets real wages fall without nominal cuts, and keeps the nominal rate off its floor. Banks are normally allowed roughly ±1 p.p. around the central number.

THE THREE PILLARS · CURRICULUM LIST

The success of an inflation-targeting regime is said to rest on exactly three things: INDEPENDENCE (free to set rates), CREDIBILITY (the market believes the target will be hit), and TRANSPARENCY (decisions explained and the bank held accountable). Lose any one and the expectations channel stops working.

Central bank	Target as stated in the curriculum	Framework
RBNZ (pioneer, 1989)	1%–3% CPI, focused on an average near 2%	Formal target
Bank of England	2% CPI within ±1.0 percentage point	Formal target
ECB	CPI close to, but below, a ceiling of 2%	Formal target
Bank of Japan	No explicit inflation measure targeted	EXCEPTION — deflation
US Federal Reserve	No formal target; statutory dual mandate	EXCEPTION — dual mandate

Source: CFA Institute, Curriculum Vol 2, LM 4 §2 "Monetary Policy Objectives", pp. 116–120, Exhibits 4 and 5. Fed and BoJ are the curriculum's two stated exceptions.

The neutral rate is trend growth plus the inflation target — and a Taylor-type rule adds a response to the inflation and output gaps on top of it

Neutral rate = real trend growth + inflation target. Above it policy is contractionary; below it, expansionary. The curriculum is candid that estimating it is more art than science.

$$i^* = r\_trend + \pi + 0.5\,(\pi - \pi^*) + 0.5\,(y - y^*) \equiv (r\_trend + \pi^*) + 1.5\,(\pi - \pi^*) + 0.5\,(y - y^*)$$

where: r_trend = real trend growth; π = observed inflation; π = target; (y − y*) = output gap as % of potential. The two forms are algebraically identical — the second makes the 1.5 loading on inflation visible.*

IMPLIED POLICY RATE — AND THE TRAP IN THE INFLATION WEIGHT

Real trend growth 2.5%, inflation target 2.0%, observed inflation 3.0%, output gap +1.0%. What rate does the rule prescribe?

1. Neutral NOMINAL rate = r_trend + π* = 2.5% + 2.0% = 4.5%. This part is straight from the curriculum.
2. Inflation gap = π − π* = +1.0%, but it is loaded 1.5× (1.0 from the π level term + 0.5 explicit) → +1.5%.
3. Output gap response = 0.5 × (+1.0%) = +0.5%.
4. i* = 4.5% + 1.5% + 0.5% = 6.50% — fully 2.0 p.p. above neutral.

Implied rate = 6.50%, i.e. 200 bp above the 4.50% neutral rate — clearly restrictive

[1] Taylor (1993) puts ACTUAL inflation in the level term plus 0.5 on the gap — total response 1.5. That is the Taylor principle.

Source: CFA Institute, Curriculum Vol 2, LM 4 §2 "The Neutral Rate", pp. 125–126, for the neutral-rate decomposition. The reaction function itself is Taylor (1993) — supplementary, not CFA Level I content.

SECTION 03

Geopolitics for investors

The curriculum frames geopolitics through four archetypes of country behaviour — autarky, hegemony, multilateralism and bilateralism

Cooperation is driven by national interest — military, economic or cultural. It is enforced through three toolkits: national security tools, economic tools and financial tools, each usable cooperatively or not.

STATE ACTORS

National governments, political leaders, central banks and finance ministries — they hold formal sovereignty over national security and national resources.

NON-STATE ACTORS

NGOs, multinationals, rating agencies, media and influential individuals. No sovereignty and no direct control of national resources — but real influence over flows and standards.

Archetype	What it describes	Curriculum examples
Autarky	Self-sufficiency; little external exchange	North Korea, Venezuela
Hegemony	A leader using its weight to control resources	China in technology transfer
Multilateralism	Mutual trade with extensive rules harmonization	Germany, Singapore
Bilateralism	Country-to-country cooperation, deal by deal	Japan, historically

Source: CFA Institute, Curriculum Vol 2, LM 5 §1 "State and Non-State Actors", pp. 144–145, and the archetype framework, pp. 167–172.

Geopolitical risk splits into three types — event, exogenous and thematic — and each one demands a completely different monitoring cadence

The Geopolitical Risk Index built by two Federal Reserve Board analysts in 2019 finds that high geopolitical risk depresses investment, employment and equity prices — and that the THREAT of an event has a larger cumulative impact than the event itself.

EVENT RISK

Built around set dates, known in advance.

Elections, referenda, new legislation, political anniversaries.

Curriculum case: UK EU referendum, 23 June 2016. GBP −8.1% day one, −11.9% at 30 days, −14.5% after a year.

Monitor with a political calendar.

EXOGENOUS RISK

Sudden and unanticipated.

Uprisings, invasions, the aftermath of natural disasters.

Curriculum case: Fukushima, 11 March 2011. Nikkei −6.2% day one, and down to −20.4% by late November.

Monitor with scenarios and hedging overlays.

THEMATIC RISK

Known, slow-moving, expanding over time.

Climate change, migration patterns, the rise of populism, terrorism, cyber threats.

No single date to trade around — the risk is a drift in the operating environment.

Monitor at strategic-review cadence.

ASSESS EVERY RISK ON THREE AXES · LIKELIHOOD · VELOCITY · SIZE AND NATURE OF IMPACT

Likelihood alone is useless — the curriculum says to weigh it only alongside velocity (how fast it hits the portfolio) and the size and nature of the impact. Because these risks rarely develop in a straight line, the recommended response is scenario building and signposting rather than a single point forecast.

Source: CFA Institute, Curriculum Vol 2, LM 5 §3 "Geopolitical Risk and the Investment Process", pp. 179–181, Exhibits 11–13; Geopolitical Risk Index, p. 187.

SECTION 04

International trade & the trade balance

Countries gain by specializing according to comparative advantage — the Ricardian insight is that absolute productivity is beside the point

Ricardo explains trade through differences in technology; Heckscher–Ohlin through differences in factor endowments — a country exports the good that uses its abundant factor intensively. Newer models add scale, variety and competition.

Specialize where your OPPORTUNITY COST is lowest → both countries gain

where: Opportunity cost of one unit of X in country A = the units of Y that A must give up to make it. Compare opportunity costs ACROSS countries — never compare output levels.

TWO COUNTRIES, TWO GOODS — ABSOLUTE ADVANTAGE IS A RED HERRING

Per hour, Home makes 10 cloth OR 5 wine. Foreign makes 4 cloth OR 4 wine. Home is more productive at BOTH goods. Should they still trade?

1. Home gives up 0.5 wine per cloth (5/10); Foreign gives up 1.0 wine per cloth (4/4).
2. Foreign gives up 1.0 cloth per wine (4/4); Home gives up 2.0 cloth per wine (10/5).
3. Comparative advantage: Home in cloth (0.5 < 1.0), Foreign in wine (1.0 < 2.0).
4. Any trade price between 0.5 and 1.0 wine per cloth — say 0.75 — leaves both strictly better off than autarky.

Specialize and trade → combined output rises even though Home is better at everything

Source: CFA Institute, Curriculum Vol 2, LM 6 §2 "Benefits and Costs of International Trade", pp. 196–198 (Ricardian and Heckscher–Ohlin models).

A tariff transfers surplus from consumers to producers and to the treasury — but consumers lose more than the other two gain, and the gap is pure deadweight loss

The welfare arithmetic: ΔCS = −(A+B+C+D), ΔPS = +A, government revenue = +C, so national welfare = −(B+D). This is the SMALL-COUNTRY case — for a large country able to move the world price, national welfare could actually rise.

Instrument	Who captures area C (the rent)	Welfare, small country
Tariff	The importing government, as tariff revenue	Falls by (B + D)
Import quota	Mixed — importer if licences sold, else foreign	Falls by (B + D), or more
Voluntary export restraint	The foreign exporter — rent leaves the country	Worse than a tariff
Export subsidy	EXPORTING country: govt spending rises	Falls (exporter loses)

SOUTH AFRICA PAPER — 20% TARIFF, FULL WELFARE DECOMPOSITION

World price USD 5.00. Domestic supply 110,000 t, demand 200,000 t → imports 90,000 t. A 20% tariff lifts the domestic price to USD 6.00: supply rises to 130,000 t, demand falls to 170,000 t, imports collapse to 40,000 t.

1. ΔCS = −(A+B+C+D) = −[USD 1 × 170,000 + ½ × USD 1 × 30,000] = −USD 185,000
2. ΔPS = +A = USD 1 × 110,000 + ½ × USD 1 × 20,000 = +USD 120,000 · Revenue = +C = USD 1 × 40,000 = +USD 40,000
3. Deadweight loss = B + D = ½ × 1 × 20,000 + ½ × 1 × 30,000 = USD 25,000

DWL = 185,000 − 120,000 − 40,000 = USD 25,000 (small-country case)

Source: CFA Institute, Curriculum Vol 2, LM 6 §3 "Trade Restrictions", pp. 200–204, Exhibits 1–3; worked Example 1 (South Africa paper), p. 202.

Trading blocs form a ladder of integration — each rung removes one more layer of friction, and the top rung gives up the currency

Forming a bloc is not automatically welfare-improving. Trade CREATION (cheaper member imports replace costly domestic output) is a gain; trade DIVERSION (member imports replace cheaper non-member imports because of the external tariff) is a loss.

Bloc type	What it adds to the rung below	Curriculum example
Free trade area	Internal barriers removed; own external policy kept	USMCA — US, CA, MX
Customs union	+ a COMMON trade policy against non-members	Benelux 1947 → EEC 1958
Common market	+ free movement of factors (labour and capital)	MERCOSUR — AR, BR, PY, UY
Economic union	+ common institutions and coordinated policy	European Union, from 1993
Monetary union	+ a single shared currency	Eurozone — 19 EU members

TRADE CREATION vs TRADE DIVERSION · THE ONLY WELFARE TEST THAT MATTERS

CREATION: a member's low-cost output displaces higher-cost domestic production inside the bloc — resources reallocate toward comparative advantage, and welfare rises. DIVERSION: a member's HIGHER-cost output displaces a cheaper non-member supplier purely because the external tariff makes the efficient supplier look expensive — welfare falls. A customs union is welfare-improving only when creation dominates diversion, which is an empirical question every single time, not a theoretical result.

Source: CFA Institute, Curriculum Vol 2, LM 6 §4 "Types of Trading Blocs", pp. 205–206; trade creation vs trade diversion, p. 206.

A trade deficit is the accounting mirror of a capital account surplus — and it is capital flows, not trade flows, that move the exchange rate in the short run

A country that imports more than it exports must borrow from foreigners or sell them assets. There is no third option — the identity is an accounting fact, not an economic theory.

$$X - M = (S - I) + (T - G)$$

where: X−M = trade balance; S = private saving; I = investment in plant and equipment; T = taxes net of transfers; G = government spending. A trade surplus therefore requires a fiscal surplus, an excess of private saving over investment, or both.

WHAT THE IDENTITY SAYS

A trade surplus means the country saves more than enough to fund its own investment; the excess buys financial claims on the rest of the world. A deficit means the reverse — it must run down those claims.

THE MIRROR IS EXACT

A trade deficit (surplus) is matched exactly by a capital account surplus (deficit). Anything that moves one moves the other by construction — you cannot analyse the trade balance in isolation.

WHAT ACTUALLY MOVES FX

Capital flows, potential and actual, are the primary driver of exchange rates in the short to intermediate term. Trade flows matter only in the long run, once prices and saving decisions have adjusted.

[1] Under BPM6 the identity is stated as current account + capital account = financial account, with net errors and omissions closing the gap.

[2] Sign conventions differ across sources — check which side the financial account carries before you net anything to zero.

Source: CFA Institute, Curriculum Vol 2, LM 7 §2 "Exchange Rates and the Trade Balance", pp. 245–246.

SECTION 05

FX market structure & quoting

FX is the largest market in the world, spot is a minority of it, and every one of the top five pairs has the US dollar on one side

The curriculum puts daily turnover at roughly USD 6.6tn for 2019 — ten to fifteen times global fixed income and about fifty times global equities. Note what dominates: FX swaps, which roll and finance positions rather than take a directional view.

Instrument	% of daily turnover
Spot	33%
Outright forwards	14%
FX and currency swaps	49%
FX options	5%

Top-5 pair	Share
USD/EUR	23.1%
JPY/USD	17.8%
USD/GBP	9.3%
USD/AUD	5.2%
CAD/USD	4.3%

WHO IS ACTUALLY TRADING · SELL SIDE, BUY SIDE, AND THE 8%

SELL SIDE: large money-centre banks acting as market makers, quoting a bid and an offer on the base currency. BUY SIDE: financial clients (real-money funds, leveraged accounts, sovereign wealth funds, central banks) and non-financial clients (corporates, retail, governments). The BIS split of daily flow is interbank 42%, financial clients 51%, non-financial clients 8% — financial-client flow overtook interbank volume for the first time in 2010. So only a small minority of FX turnover is anyone actually paying for goods. London is the largest trading hub, then New York, then Tokyo; liquidity is deepest between roughly 08:00 and 11:30 New York time, when London and New York are both open.

Source: CFA Institute, Curriculum Vol 2, LM 7 §2, p. 216 and Exhibits 2–4, pp. 227–228 (BIS Triennial Survey, April 2016). The 2022 BIS survey put daily turnover near USD 7.5tn.

Every FX quote reads A/B: the number of units of A, the PRICE currency, that buys one unit of B, the BASE currency

Hierarchy for the base currency: EUR outranks everything; then GBP where there is no euro; then USD where there is neither. AUD and NZD are the exceptions — they are the base against the dollar (USD/AUD, USD/NZD).

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A / B   =   units of A (PRICE currency)  per  ONE unit of B (BASE currency)
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where: The base is always quantity one. Direct quote = your domestic currency is the price currency; indirect = it is the base. Most majors are quoted to 4 decimal places; the yen is the exception, at 2.

Market name	Actual ratio quoted	How to read it out loud
Euro	USD/EUR = 1.1701	EUR is the base — one euro costs USD 1.1701
Sterling	USD/GBP = 1.3118	GBP is the base — one pound costs USD 1.3118
Dollar–yen	JPY/USD = 111.94	USD is the base — one dollar costs JPY 111.94
Dollar–Canada	CAD/USD = 1.3020	USD is the base — one dollar costs CAD 1.3020
Euro–sterling	GBP/EUR = 0.8920	EUR is the base — one euro costs GBP 0.8920

PERCENTAGE MOVES ARE NEVER SYMMETRIC

USD/EUR rising 1.1500 → 1.2000 is a +4.35% appreciation of the EURO (the base). The matching dollar move is 1.1500/1.2000 − 1 = −4.17%, not −4.35%. Mathematically these two can never be equal. Always compute the percentage change on the currency that is the BASE of the quote you were handed — then invert if you were asked about the other one.

Source: CFA Institute, Curriculum Vol 2, LM 7 §2 "Exchange Rate Quotations", pp. 229–232, Exhibit 5; percentage-change asymmetry, p. 232.

A two-sided price is always about buying and selling the BASE currency — and inverting a quote swaps the bid and the offer

The bank buys the base low and sells it high. The bid is therefore always below the offer — in EVERY quote, in EVERY direction. If your inversion breaks that, you inverted it wrong.

MXN/USD TWO-SIDED PRICE — AND THE INVERSION TRAP

A New York dealer quotes 18.8580 – 18.8600 MXN/USD to a Mexico City client. Interpret it, then produce the inverted USD/MXN two-sided price.

1. Base = USD, price = MXN. Both sides of the quote refer to the dealer buying or selling USD.
2. Dealer BUYS USD at its bid, 18.8580 (client sells USD); dealer SELLS USD at its offer, 18.8600. Spread = 20 pips.
3. To invert you must FLIP AND CROSS: bid(USD/MXN) = 1 / offer(MXN/USD) = 1/18.8600; offer(USD/MXN) = 1 / bid = 1/18.8580.

USD/MXN = 0.053022 – 0.053028 · bid still below offer, as it must be

WHY A SPREAD EXISTS AT ALL · THREE FRICTIONS THE DEALER IS PRICING

INVENTORY RISK — the dealer holds the position while looking for the other side, and the market can move meanwhile. ADVERSE SELECTION — better-informed flow picks off a stale quote, so the dealer widens against the risk of being the last to know. OPERATIONAL AND CREDIT COST — settlement, capital, and counterparty lines all have to be paid for. Electronic dealing has compressed major-pair spreads to a fraction of a pip; emerging-market pairs remain an order of magnitude wider, and "spreading the client" is still how a dealer defends a margin.

Source: CFA Institute, Curriculum Vol 2, LM 7 §2 "Exchange Rate Quotations", pp. 229–231; Example 3 (MXN/USD), p. 232.

The IMF sorts exchange-rate regimes into eight categories, and where a country sits tells you exactly how much monetary autonomy it has surrendered

Regimes are read most-fixed at the top to most-flexible at the bottom. The curriculum warns the boundaries are somewhat arbitrary, they shift over time, and even an "independent float" gets intervened in occasionally.

THERE CAN BE NO IDEAL CURRENCY REGIME · PICK TWO OF THREE

The ideal regime would need three things at once: exchange rates CREDIBLY FIXED between all currencies, every currency FULLY CONVERTIBLE, and each country free to run an INDEPENDENT MONETARY POLICY for its own domestic objectives. The curriculum is blunt — the third is impossible alongside the first two, because capital simply arbitrages away any rate the central bank tries to set independently. Every real regime below is a choice about which one to give up.

IMF regime (fixed → flexible)	What it means	Curriculum examples
1 · No separate legal tender	Dollarization, or a shared-currency union	Panama, Ecuador · the EMU
2 · Currency board	100% FX reserve backing of the base money	Hong Kong SAR · Bulgaria
3 · Fixed parity	Pegged to an anchor, narrow defended band	Saudi Arabia, Qatar, Denmark
4 · Target zone	Wider band around a central parity	Slovak Republic (pre-euro)
5 · Crawling peg	Parity reset on a pre-announced schedule	Nicaragua · Botswana
6 · Crawling band	Band around a parity that itself moves	China, Croatia, Switzerland
7 · Managed float	Intervention, but no announced target	Malaysia, Russia, Czech Rep.
8 · Independent float	Market-determined; rare intervention	US, UK, Japan, Canada, Chile

Source: CFA Institute, Curriculum Vol 2, LM 7 §3 "The Ideal Currency Regime", p. 233, and Exhibit 6 "A Taxonomy of Currency Regimes" (IMF classification as of 30 April 2008), pp. 237–238.

SECTION 06

FX arithmetic — cross-rates, IRP, PPP

Real exchange rates deflate the nominal quote by relative price levels — they are an analyst's index of purchasing power, never a tradeable price

Stated with the DOMESTIC currency as the price currency, the real exchange rate is simply the real price you face for a foreign basket. The higher it goes, the less your purchasing power abroad. And it is an index — nobody trades it.

$$R_{d/f} = S_{d/f} \times (P_f / P_d) \qquad \%\Delta R \approx \%\Delta S + \%\Delta P_f - \%\Delta P_d$$

where: $S_{d/f}$ = nominal spot, domestic per one foreign unit; P_f, P_d = foreign and domestic price levels. The right-hand expression is the additive APPROXIMATION of the exact product form.

A UK CONSUMER BUYING EUROZONE GOODS

Over one year the nominal GBP/EUR rate rises 10%, Eurozone prices rise 5%, and UK prices rise 2%. What happens to the real cost of the same Eurozone basket for a UK buyer?

1. Note the convention: in GBP/EUR the UK's own currency is the PRICE currency, not the base. That is what makes the formula work.
2. Exact: (1 + 0.10) × (1.05 / 1.02) − 1 = +13.24%.
3. Approximation: +10% + 5% − 2% ≈ +13%. Close, because the cross terms are small at these magnitudes.
4. A HIGHER real exchange rate means a higher real price for the foreign basket — so UK purchasing power over Eurozone goods FELL by about 13%.

ΔR ≈ +13% (13.24% exact) → UK real purchasing power over EZ goods down ~13%

Source: CFA Institute, Curriculum Vol 2, LM 7 §2 "The Foreign Exchange Market and Exchange Rates" — real exchange rates, pp. 218–220.

A cross-rate is built by chaining two dollar quotes so the dollars cancel — and when the dollar sits on the same side of both, you must invert one first

Write the units before you touch the calculator. If the shared currency does not cancel, you are about to get the wrong answer — invert a leg and try again.

$$\mathrm{CAD/EUR} = (\mathrm{CAD/USD}) \times (\mathrm{USD/EUR}) \qquad \mathrm{JPY/CAD} = (\mathrm{CAD/USD})^{-1} \times (\mathrm{JPY/USD})$$

where: Left: USD is the base of one quote and the price of the other, so it cancels directly. Right: USD is the BASE of both quotes, so one must be inverted before anything cancels.

BOTH CASES FROM ONE SET OF QUOTES

Market quotes: USD/EUR = 1.1701, CAD/USD = 1.3020, JPY/USD = 111.94. Build the euro–Canada and the Canada–yen crosses.

1. CAD/EUR — chain EUR → USD → CAD: one euro buys USD 1.1701, and each of those buys CAD 1.3020.
2. CAD/EUR = 1.3020 × 1.1701 = 1.5235 CAD per EUR. (CAD)/(USD) × (USD)/(EUR): the USD cancels.
3. JPY/CAD — USD is the BASE in both quotes, so nothing cancels. Invert one: 1 / 1.3020 = 0.7680 USD/CAD.
4. JPY/CAD = 0.7680 × 111.94 = 85.97 JPY per CAD.

CAD/EUR = 1.5235 · JPY/CAD = 85.97 (yen to 2 dp, the rest to 4)

Source: CFA Institute, Curriculum Vol 2, LM 8 §1 "Cross-Rate Calculations", pp. 254–255.

When a dealer's cross-rate drifts from the chain-implied value, three trades close the triangle and lock in a riskless profit

The curriculum's own illustration: chain-implied JPY/CAD of 85.97 against a misquote of 86.20 gives a riskless JPY 0.23 per CAD 1. Buy the currency cheap in one venue, sell it dear in the other, close the triangle.

PRACTICE PROBLEM — ZAR/SEK ACROSS TWO DEALERS

Dealer A quotes CNY/ZAR = 0.9149 and CNY/SEK = 1.0218. Dealer B quotes the ZAR/SEK cross directly at 1.1210. Construct the trade on SEK 1 million.

1. Chain-implied cross via Dealer A: ZAR/SEK = (CNY/SEK) / (CNY/ZAR) = 1.0218 / 0.9149 = 1.1168.
2. Dealer B is quoting 1.1210 — higher, so B pays MORE rand for each krona than A charges.
3. Buy SEK from A: pay ZAR 1,116,800 for SEK 1,000,000. Sell that SEK to B: receive ZAR 1,121,000.
4. Profit = SEK 1,000,000 × (1.1210 − 1.1168) = ZAR 4,200.

ZAR 4,200 per SEK 1 million traded — riskless, and no capital committed

THE DISTRACTOR TO WATCH · PER MILLION OF WHICH CURRENCY?

The profit is ZAR 4,200 per million SEK traded — NOT SEK 4,200, and not per million rand. The rate is quoted in rand per krona, so the notional is in krona and the profit lands in rand. Every one of the wrong answers in the curriculum's question set is built on getting exactly that pairing backwards. Method, every time: chain the liquid quotes, compare to the direct cross, and trade only if the gap survives round-trip transaction costs.

Source: CFA Institute, Curriculum Vol 2, LM 8 §1 — the JPY/CAD triangular-arbitrage illustration, p. 255, and Practice Problems 1–2, pp. 266–268.

The forward rate is pinned by a no-arbitrage condition between two equivalent investments — this is covered interest rate parity, and it is not a forecast

Forward points are just (F − S) rescaled to the last decimal of the spot quote: × 10,000 for non-yen majors quoted to 4 dp, × 100 for the yen at 2 dp. Day count on the deposit legs is Actual/360.

$$F_{f/d} = S_{f/d} \times (1 + r_f \tau) / (1 + r_d \tau) \qquad F_{f/d} - S_{f/d} = S_{f/d} \times [(r_f - r_d) / (1 + r_d \tau)] \times \tau$$

where: Quoted f/d: units of foreign per ONE domestic unit, so the DOMESTIC currency is the base. τ = actual days / 360. Points are proportional to spot and to the rate differential, and only approximately proportional to tenor.

Rates (f/d quote: base = d)	Forward points	The BASE currency (d) is trading at a…
$r_f > r_d$ — base rate is LOWER	Positive: F > S	PREMIUM — foreign currency at a discount
$r_f < r_d$ — base rate is HIGHER	Negative: F < S	DISCOUNT — foreign currency at a premium
$r_f = r_d$	Flat: F = S	Neither — no interest wedge to arbitrage

THE ONE RULE THAT ANSWERS MOST FX QUESTIONS

Regardless of quoting convention, the currency with the HIGHER interest rate always trades at a forward DISCOUNT. Read the quote, identify the base, compare the two rates — and you already know the sign of the points before doing any arithmetic. Use the calculation to get the magnitude, and this rule to check you have not inverted it.

Source: CFA Institute, Curriculum Vol 2, LM 8 §2 "Forward Rate Calculations" — arbitrage relationships and forward discounts/premiums, pp. 259–264.

When a dealer misquotes the forward, the arbitrage direction is set by where the mispricing sits — not by which currency pays the higher rate

Read the curriculum's own conclusion carefully: the investor here BORROWS at the higher of the two rates and still profits, because the foreign currency is underpriced in the forward market.

MISQUOTED 12-MONTH FORWARD — FIND THE 77 BASIS POINTS

Spot $S_{f/d} = 1.6535$. Domestic 12-month rate $r_d = 3.50\%$, foreign 12-month rate $r_f = 5.00\%$. A dealer quotes the 12-month forward at $F = 1.6900$.

1. No-arbitrage forward: $F_{f/d} = 1.6535 \times (1.0500 / 1.0350) = 1.6775$.
2. The quote of 1.6900 is too HIGH — it takes more foreign currency per domestic unit forward, so the foreign currency is underpriced forward.
3. Return on the hedged foreign route: $S(1 + r_f) \times (1/F) = 1.6535 \times 1.05 \times (1/1.6900) = 1.0273$, i.e. 2.73%.
4. The domestic-only route returns 3.50%. So: borrow foreign at 5%, sell it spot, lend domestic at 3.50%, buy the foreign back forward at 1.6900.

Riskless spread = 3.50% − 2.73% = 77 bp per unit, with no capital committed

WHY "BORROW LOW, LEND HIGH" IS THE WRONG INSTINCT HERE

Look at what the trade actually does: it borrows at the HIGHER rate (5% foreign) and lends at the LOWER (3.50% domestic) — and still profits, because the mispricing is in the forward, not in the rates. Borrow-low-lend-high is the CARRY trade, which is unhedged and carries FX risk. This is arbitrage: fully hedged, and its direction is set purely by which side of the no-arbitrage forward the dealer has quoted. Compute F first, compare, then decide the direction.

Source: CFA Institute, Curriculum Vol 2, LM 8 §2 — the misquoted-forward arbitrage worked example, pp. 260–261.

Purchasing power parity says identical goods should cost the same everywhere — the curriculum is unusually blunt about how weakly that holds

The Economist's Big Mac index takes a single standardised good and still shows wide international dispersion — the curriculum uses it precisely to make the point that this dispersion is typical of most goods and services.

$$\text{Absolute PPP: } S_{d/f} = P_d / P_f \qquad \text{Relative PPP: } \%\Delta S_{d/f} \approx \pi_d - \pi_f$$

where: Absolute PPP needs identical baskets and no frictions, and pins the LEVEL. Relative PPP only constrains the CHANGE — it can never tell you whether today's level is right.

WHY IT FAILS — THE CURRICULUM'S OWN LIST

- Goods and services are not identical across countries
- Countries produce and consume genuinely different baskets
- Many goods and services are never traded internationally at all
- Trade barriers and transaction costs — shipping, import taxes
- Capital flows are at least as important as trade flows in setting nominal FX

SO WHAT IS IT STILL GOOD FOR?

- Be honest about the verdict: the text says relative purchasing power shows a weak — if any — tendency to equalise over the long run
- Under high inflation, relative PPP does pin the nominal rate tightly
- A first-pass fair-value screen on EM currencies
- Big Mac / iPhone indices: communication devices, not models

Source: CFA Institute, Curriculum Vol 2, LM 7 §2 — purchasing power parity and the Big Mac index, pp. 218–219.

Drill the forward calculation until the sign is automatic — the base currency with the higher interest rate always trades at a discount

Every quote below is written price/base. Cover the last two columns, work each row, then check both the number AND the sign of the points against the rule.

Spot (price / base)	r on price ccy	r on base ccy	Forward 12m	Points
USD/EUR = 1.1000	5.00% (USD)	2.00% (EUR)	1.1000 × 1.05/1.02 = 1.1324	+323.5 prem
GBP/EUR = 0.8500	4.00% (GBP)	3.00% (EUR)	0.8500 × 1.04/1.03 = 0.8583	+82.5 prem
JPY/USD = 150.00	0.25% (JPY)	5.00% (USD)	150.00 × 1.0025/1.05 = 143.21	−678.6 disc
CHF/EUR = 0.9500	1.50% (CHF)	3.50% (EUR)	0.9500 × 1.015/1.035 = 0.9316	−183.6 disc

FOUR-LINE FORWARD CHEATSHEET · WORKS ON EVERY QUESTION

1. Find the BASE — it is the second currency in the A/B label, and it is always quantity one.
2. Compare the two rates. Base rate HIGHER ⇒base at a forward DISCOUNT (F < S). Base rate LOWER ⇒PREMIUM. You now know the sign.
3. Compute F = S × (1 + r_price·τ) / (1 + r_base·τ). The PRICE currency's rate goes on top — this is the step everyone inverts.
4. Points = (F − S) × 10,000, or × 100 for the yen. Check the sign against step 2 before you write anything down.

[1] Drills use annual rates with τ = 1, so the day-count term drops out of both legs of the ratio.
[2] For any other tenor use F = S × (1 + r_price·τ) / (1 + r_base·τ), with τ = actual days / 360.

Source: CFA Institute, Curriculum Vol 2, LM 8 §2 "Forward Rate Calculations" — arbitrage relationships and points scaling, pp. 259–264.

KEY TAKEAWAYS

Deck 5 — Macro Policy, Geopolitics, Trade & FX

- Fiscal multiplier = $1/[1 - c(1 - t)]$. At $c = 0.9$, $t = 0.2$ it is 3.57 — the tax rate is the leakage that cuts it from 10. Never confuse it with the MONEY multiplier.

- Debt is sustainable while real g > real r. Japan carries 237% of GDP and pays 0.4% of GDP in net interest — the rate matters more than the stock.

- Three monetary tools (OMO, policy rate, reserve requirements); four transmission channels (short rates, asset prices, FX, expectations); neutral rate = trend growth + inflation target.

- Inflation targeting rests on independence, credibility and transparency. The curriculum's two named exceptions are the Bank of Japan and the Federal Reserve.

- Comparative advantage is about opportunity cost, not output. A tariff's deadweight loss = $|\Delta CS| - \Delta PS -$ revenue — in the South Africa case, 185 − 120 − 40 = USD 25,000.

- A trade deficit is the exact mirror of a capital account surplus. Capital flows drive FX in the short and intermediate run; trade flows only in the long run.

- Read every quote as A/B = units of A (price) per ONE unit of B (base). Inverting a two-sided quote flips AND crosses: new bid = 1/old offer.

- Covered IRP: $F = S \times (1 + r_{price} \cdot \tau)/(1 + r_{base} \cdot \tau)$. The higher-rate currency always trades at a forward DISCOUNT — but arbitrage DIRECTION comes from the forward quote, not the rate ranking.